

Physical Activity During School Recess

A Systematic Review

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Context: Interest has increased in examining the physical activity levels of young people during school recess. Identifying correlates of their recess physical activity behaviors is timely, and would inform school-based physical activity programming and intervention development. The review examined the correlates of children's and adolescent's physical activity during school recess periods.

Evidence acquisition: A systematic search of six electronic databases, reference lists, and personal archives identified 53 studies (47 focused on children) published between January 1990 and April 2011 that met the inclusion criteria. Data were analyzed in 2011. Correlates were categorized using the social-ecological framework.

Evidence synthesis: Forty-four variables were identified across the four levels of the social-ecological framework, although few correlates were studied repeatedly at each level. Positive associations were found of overall facility provision, unfixed equipment, and perceived encouragement with recess physical activity. Results revealed that boys were more active than girls.

Conclusions: Providing access to school facilities, providing unfixed equipment, and identifying ways to promote encouragement for physical activity have the potential to inform strategies to increase physical activity levels during recess periods.

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Context

Physical activity is positively associated with psychological well-being, bone health, and motor skill development and negatively associated with waist circumference and clustering of cardiovascular disease risk factors.^{1–4} However, as children and adolescents may not be engaging in sufficient physical activity to benefit their health,¹ the promotion of physical activity is a public health priority.⁵

Children and adolescents spend a substantial proportion of their waking hours at school. Noncurricular time, such as school recess periods (recess and lunchtime) and afterschool programs, provide opportunities for children to be physically active within the school environment.⁶ Of these contexts, recess periods may provide the single

greatest opportunity during the school year to affect physical activity levels.⁷ However, a recent trend has been to reduce the frequency and duration of school recess, or remove it altogether, often because of academic pressures. Consequently, it is important that recess be included in school-based physical activity programming and policy and that the recess environment is conducive for youth to make physically active choices.⁸ Although the scheduling and duration of recess vary among countries, schools should provide social and physical environments that facilitate enjoyable and safe physical activity engagement in this context.⁹

A number of reviews^{10–12} have examined correlates of preschool, children's and adolescents' physical activity, yet these typically have focused on factors associated with whole-day activity. Because physical activity is a multidimensional behavior influenced by numerous factors across several domains,¹⁰ it is logical to also consider specific contexts in which children and adolescents are active. Few reviews have focused on the correlates of specific parts of the school day, such as school recess periods, yet synthesizing such information could help to inform both school-based physical activity programming and intervention development aimed at increasing children's physical activity levels. Further, this information

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potentially can inform future policy concerning recess periods and their role within the school day.

The aim of this review was to examine correlates of children's and adolescents' physical activity during school recess periods. Hereafter, physical activity during school recess is referred to as recess physical activity. For the purpose of this review, school recess was defined as the noncurricular time allocated by schools between classes for youth to engage in leisure activities.¹³

Evidence Acquisition

A systematic literature search of papers was conducted in six electronic databases (PubMed, SportsDiscus, Web of Science, ProQuest, Cochrane, and Scopus). Search strategies for the various databases included the following key words in three main areas: population (*child, infant, youth, adolescent*); school (*school, primary, elementary, middle school, high school, secondary school*); and recess (*breaktime, break time, school recess, recess, playtime, lunchtime, free play*). Only papers that had been published in peer-reviewed journals were considered for inclusion. Abstracts, expert opinion, case studies, conference proceedings, and dissertations were not included. Bibliographies of retrieved studies and the authors' personal collections also were searched.

For inclusion in the review, studies were required to (1) include participants aged 5–18 years; (2) have a measure of physical activity as the outcome variable; (3) specifically examine physical activity during recess and/or lunchtime; (4) examine associations between physical activity and other variables; (5) be published between January 1990 and April 2011; and (6) be published in peer-reviewed journals in English. School recess periods included morning, lunchtime, and afternoon. A previous review¹³ of recess physical activity levels reported that only two studies were published before 1990; this was used as the start date for the current review. Data were collected and analyzed in 2011.

Studies that reported total daily recess (e.g., sum of morning recess and lunchtime) or investigated specific recess periods were included. Before-school time and afterschool programs conducted on school sites were not considered recess periods in this review. Studies that reported findings before/after school time and during school time were included where the results were presented separately for the recess periods. Data from control groups, baseline intervention studies, and intervention studies where the correlate was reported separately as a confounding variable were included. Longitudinal studies were included if at least one measurement period was conducted during elementary school and results at this time point reported. Studies that reported only the effectiveness of recess-based interventions on physical activity levels were excluded.

Studies that investigated physical activity levels during recess periods in special populations (e.g., autistic spectrum disorders, attention deficit disorders) also were included in the review, as these studies collected data from schools where children with and without particular disorders had access to the same school environment and had recess and/or lunchtime together. When comparisons were made between children with and without

particular disorders and no differences were found, all data were included for review. In circumstances where differences occurred, data from the children without the disorders were considered to enable comparisons with other studies. Lastly, studies that investigated physical activity levels solely of special populations during recess periods were included but noted accordingly in the results.

Results obtained from the initial literature search were assessed independently. Potentially relevant papers were selected by screening the (1) title; (2) abstract; and (3) full paper if the abstract provided insufficient detail. Data extraction was then undertaken using standardized forms and included country of study; study design; sample characteristics (e.g., sample size, age, gender); the physical activity measure used and the outcome variable; recess period(s) studied; recess characteristics (e.g., duration, supervision, environment); and results. Appendix A (available online at www.ajpmonline.org) provides an overview of the characteristics of the studies that met the inclusion criteria. Data are presented separately for children (aged 5–12 years) and adolescents (aged 13–18 years). Because of cross-study variability in the age at which children transition into high schools, if children were aged 11–12 years but the study was conducted in secondary schools, they were classed as adolescents.

Categorization of Variables

For consistency and comparability with previous reviews of correlates of youth physical activity,^{10–12} a social–ecological framework was used. This model posits that multiple levels of influence interact to promote or constrain participation in physical activity.¹⁴ Identified variables were categorized into four groups according to the framework; individual, social, physical environment, and organizational/policy. Conceptually similar variables were combined (e.g., different facilities available for use during recess periods) to enable consistency with interpretation.

A range of measures were used to examine recess physical activity (Appendix A, available online at www.ajpmonline.org). Children's physical activity largely was measured using objective measures, with accelerometry the most commonly used method (36% of all studies). Adolescents' physical activity typically was measured using subjective measures (e.g., self-report questionnaires; 71% of adolescent studies). Multiple physical activity measures were used in six children's studies and one adolescent study, respectively. When moderate (MPA) and vigorous (VPA) physical activity were reported separately and the associations obtained were in different directions, they were treated as separate results and noted accordingly.

If the MPA and VPA results were reported in the same direction, the results were combined to form one overall result (i.e., MVPA). For studies that used two or more objective measures, a combined result was reported when results from both of these measures were in the same direction.¹⁰ Where differences in the direction of the association occurred between measures, the results were presented separately and noted accordingly. The same approach was adopted when studies examined school recess periods separately (e.g., morning recess and lunchtime) and reported different findings. A variety of techniques were used to evaluate associations (e.g., cor-

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issue.

Table 1. Rules for classifying variables regarding strength of associations with recess physical activity¹¹

Studies supporting association (%)	Summary code	Explanation of code
0–33	0	No association
34–59	?	Indeterminate/inconclusive association
60–100	+	Positive association
60–100	–	Negative association

Note: When an outcome had been studied four or more times, it was coded as 00 (no association); ?? (indeterminate); ++ (positive association); or -- (negative association)

relation, multilevel modeling, ANOVA). Several studies reported univariate analyses, although some also reported multivariate analyses with adjustments for confounding variables. Where possible, the findings reported are from the fully adjusted models.

The results were coded using the procedure outlined in previous reviews where the association between an identified variable and recess physical activity was determined by the number of findings that support the direction of association.^{2,10,11} The summary column identifies the number of studies finding positive (+); negative (–); or no (0) associations between recess physical activity and the identified variable. The left column indicates the percentage of findings that demonstrated an association between recess physical activity and the identified variable. When 0%–33% of the findings supported an association, the result was classified as no association (0). An indeterminate/inconclusive (?) classification was determined if 34%–59% of the findings supported an association, whereas a positive or negative association was determined if 60%–100% of the findings supported the direction of the association (Table 1). If a variable was investigated four or more times and supported an association as positive, the result was coded ++, while negative associations were coded -- and no associations coded as 00.

Evidence Synthesis

Of the 2718 studies identified, 53 papers were included in the review (Figure 1). The majority of studies were cross-sectional ($n=42$); focused on children ($n=47$); and reported MVPA as the outcome variable ($n=26$). Only one study¹⁵ included children and adolescents and reported the results separately. Eight studies used self-report measures to assess physical activity, with seven and five of these reporting the validity and/or reliability of the measures used, respectively. Because of the limited number of adolescent studies, data from children and adolescents are reported together to obtain the summary coding variables reported below.

The majority of studies were conducted in the U.S. ($n=16$)^{16–31} or the United Kingdom and Northern Ireland ($n=12$).^{32–43} Five studies were conducted in Australia,^{44–48} four in Norway,^{15,49–51} three in New

Zealand,^{52–54} two each in Taiwan,^{55,56} Japan,^{57,58} and Portugal^{59,60}; one each in Hong Kong,⁶¹ Mexico,⁶² the Philippines,⁶³ Canada,⁴⁴ Belgium,⁶⁴ Hungary⁶⁵, Spain,⁶⁶ and France.⁶⁷ Only one study collected data across multiple countries.⁴⁴ Four studies collected data from morning recess,^{17,30,58,66} seven from lunchtime,^{23,32,57,52,53,63,67} and 12 from daily recess (i.e., no time of day reported).^{16,22,24,26–28,46,49–51,61,62} The remaining 30 studies collected data across multiple recess periods (e.g., morning recess and lunchtime).

Correlates of School Recess Physical Activity

Table 2 summarizes associations between identified correlates and recess physical activity. Correlates that have been examined four or more times are reported below.

Individual Variables

Sixteen individual variables were identified from 47 studies, with five of these being investigated four or more times. The most frequently studied variable was gender (38 studies), with boys consistently found to be more active than girls.^{15–18,21–23,25,27–32,34–35,37–43,45–49,51–52,54,59–60,62–66} No associations were found for grade level^{15,18,20,23} or BMI/central adiposity^{18,30,39,40,42,54}; evidence for differences in physical activity between children with and without special education needs^{21,27,55–58} and age^{17,30,33,34,35,37,39,52,54,55,59,63,67} were inconclusive.

Social Variables

Associations between social variables and physical activity were examined in ten studies. Five social variables were identified, with associations of perceived encouragement, SES, and supervision with physical activity investigated four or more times. A positive association was found between physical activity and perceived encouragement from friends, family, and schools.⁵³ Higher levels of perceived encouragement were associated with higher physical activity levels.⁵³ There was inconclusive

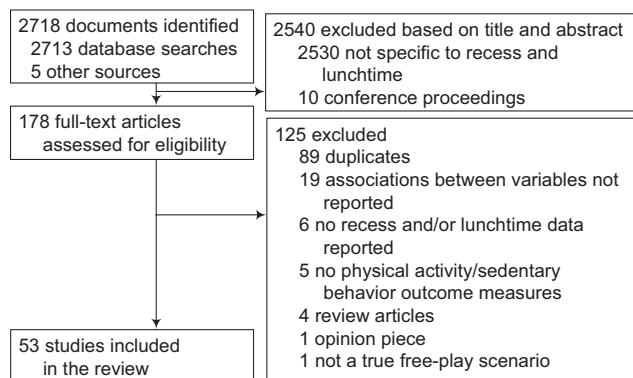
**Figure 1.** Flow diagram of search results published between January 1990 and April 2011

Table 2. Summary of reported outcomes of studies published January 1990 to April 2011

Correlate	Found association with physical activity during recess (references)	Association (±)	Found no association with physical activity during recess (references)	Summary coding (n/N for row [%])	Summary code for association (–/+)
INDIVIDUAL VARIABLES					
Behavioral					
Day-to-day variability			36	0/1 (0)	0
Physical conflict frequency	40	+		1/1 (100)	+
Sedentary activities (time)	40	–		1/1 (100)	–
Play ball games	46	+		1/1 (100)	+
Sports activities (time)	40	+		1/1 (100)	+
Biological					
Age	30; 33 ^b ; 39; 59; 67 33 ^a , 54, 55 ^c , 63	– +	17, 34, 35, 37, 52, 55	5/15 (33)	00
Grade level	15 18	– +	20, 23	1/4 (25)	00
BMI (overweight)	18 ^d , 39 ^{b,d} , 54 ^c 54 ^m	– +	18 ^c , 30, 39 ^c , 40, 42, 54 ^d	3/10 (30)	00
Body mass	67	–		1/1	+
Maturation	67	–		1/1	+
Ethnicity (Caucasian)	54	–	18	1/2 (50)	?
Fitness			20	0/1 (0)	0
Gender (male)	15–18, 21, 23, 25, 28, 29 ^{step} , 31, 32, 34, 35, 37, 38 ^b , 39, 40, 42, 43 ^{a,b,d} , 45, 46, 47 ^b , 48, 51, 52, 54, 59, 64, 65, 66 60	+ –	22, 27, 29 ^{HR} , 30, 35 ^b , 38 ^a , 43 ^{b,c} , 47 ^a , 49, 62, 63, 66	32/43 (74)	++
Special educational needs (no disability)	21, 55, 56, 58 ^{PDD}	+	20, 27, 57, 58 ^{ADHD}	4/8 (50)	??
Cognitive					
Interest in physical activity	50, 51	+		2/2 (100)	+
Enjoyment			29	0/1 (0)	0
Social					
Group size (10 or more)	40	+		1/1 (100)	+
Perceived encouragement (including from friends, parents, school, and teachers)	53 ^{d,e,h} , 53 ^{d,e,i} , 53 ^{d,f,h} , 53 ^{d,g,h}	+	53 ^{d,f,i} , 53 ^{d,g,i}	4/6 (67)	++
SES (low)	61, 63 ^d	–	45, 63 ^c	2/4 (50)	??
Supervision	25; 46 (observation, presence)	–	38, 40, 47 ^{a,d} ; 48	3/8 (38)	??
Supervision	47 ^{b,d}	+			
Teacher management			46, 48	0/2 (0)	0
PHYSICAL ENVIRONMENT					
Aesthetics			32	0/1 (0)	0
Unfixed equipment	25; 38 ^a ; 40; 46; 47 ^b ; 48 (ball/child ratio) ^b	+	38 ^b ; 46 (no. of balls); 47 ^a , 48	6/10 (60)	++

(continued on next page)

Table 2. Summary of reported outcomes of studies published January 1990 to April 2011 (continued)

Correlate	Found association with physical activity during recess (references)	Association (±)	Found no association with physical activity during recess (references)	Summary coding (n/N for row [%])	Summary code for association (–/+)
Facility availability and accessibility					
Outdoor physical activity areas	15 (sledding hill, soccer field ^d); 50 (green space)	+	15 (ball areas, green spaces, soccer field ^k); 47 (sports fields); 50 (ball areas, soccer fields, ski, water, woods)	3/12 (25)	00
Outdoor space	38 (space/child) ^b	+	38 (space/child ^a , total space); 46 (total space)	1/4 (25)	00
Indoor physical activity spaces/areas	50 (gym)	+	50 (ice skating, swimming)	1/3 (33)	0
Fixed equipment/markings	15 (equipment ^l , hopscotch ^l); 44 ^b ; 46 (markings, shading); 47 ^a ; 50 (equipment, obstacle course)	+	15 (boarding area, fixed equipment ^k , hopscotch ^k , obstacle course); 32 (design); 38 (fixed equipment, markings); 50 (boarding area, climbing, fenced court yard)	8/19 (42)	??
	47 ^b	–			
Overall facility provision	15 ^h ; 32 (sports provision); 50; 51	+	15 ⁱ ; 32 (facility provision)	4/6 (67)	++
Play location (indoor)	19	–	65	1/2 (50)	?
Play surface (grass)	44, 46	+	47	2/3 (67)	+
School location (rural)	23	+		1/1 (100)	+
Seasons/temperature/weather	38 (temp) ^b ; 40 (temp); 46 (raining, temp)	–	34 (season); 36 (season); 38 (temp) ^a ; 46 (humidity); 48 (heat stress)	4/9 (44)	??
ORGANIZATIONAL/POLICY					
Written policy	51	+		1/1 (100)	+
Physical activity program involvement			51	0/1 (0)	0
Organized activities	51	+		1/2 (50)	?
	25	–			
Number on roll	48	–	38	1/2 (50)	?
Recess duration	39 ^{b,c} , HR ⁱ ; 45	+	37 ^a , 39 ^{a,c} , 39 ^{a,b,d}	3/7 (43)	??
	24, 39 ^{b,c} , ACC ^j ; 46	–			
Class time vs recess	22	–		1/1 (100)	–
Fitness break vs recess	29	+		1/1 (100)	+
Morning vs lunchtime recess	25	+		1/2 (50)	?
	48	–			
Physical education vs recess	20, 26 ^b , 41, 61	–		4/7 (57)	??
	26 ^a , 28 ⁿ , 56, 62	+	28 ^o		
Provision of physical education (5 times/week—yes)			51	0/1 (0)	0
Number of daily recess periods			34	0/1 (0)	0

Note: The summary code is an overall summary of the findings for each variable. 0, no association; ?, indeterminate/inconclusive association; ??, indeterminate/inconclusive association for an outcome that has been studied four or more times; +, positive association; ++, positive association for an outcome that has been studied four or more times; –, negative association. “Supervision” indicates children were supervised by adults (e.g., lunchtime assistants) and teachers, but no direct management or organization of games/activities during recess/lunch. For references 22, 26, and 61, data were collected from special population only. For reference 55, data are from children without disorders although study included special populations.

^aPAssociation for: ^aMPA, ^bVPA, ^crecess, ^dlunchtime, ^efriend support, ^fparent support, ^gschool support, ^hjunior high adolescents, ⁱsenior high adolescents, ^jmale adolescents, ^kfemale adolescents, ^lprimary school children, ^mcentral adiposity, ⁿgirls, ^oboys

ACC = accelerometry; ADHD = attention-deficit/hyperactivity disorder; HR, heart rate; MPA, moderate physical activity; PDD = pervasive developmental disorder; Step = pedometer; VPA, vigorous physical activity

evidence for associations of SES^{45,61,63} and adult supervision^{25,38,40,46–48} with physical activity.

Physical Environmental Variables

Twelve physical environmental variables were identified from 16 studies, six of which were studied four or more times. The most frequently studied variables were availability of facilities in the outdoor school grounds, and fixed equipment and markings within the school grounds. The availability of separate facilities in outdoor physical activity spaces/areas (e.g., ball areas, sports fields, green spaces) has been investigated mainly in adolescents, and no association with physical activity was found.^{15,47,50} However, when overall facility provision was considered (i.e., the sum of facilities available), positive associations with physical activity were identified.^{15,32,50,51} Fixed equipment and markings have been investigated in children and adolescents, and the association with physical activity was inconclusive.^{15,32,38,44,46,47,50} In comparison, unfixed equipment (e.g., loose equipment, balls, skipping ropes) has been investigated only in children, with positive associations found.^{25,38,40,46–48} An indeterminate result was obtained for weather/seasonal differences.^{34,36,38,40,46,48} No association was found between available outdoor space and physical activity.^{38,46}

Organizational/Policy Variables

Eleven variables were identified from 19 studies, two of which had been studied four or more times. An inconclusive association was found between recess duration (which typically is determined by school policy) and physical activity.^{24,37,39,45,46} Differences in physical activity levels between recess periods and physical education have been investigated primarily in children with special education needs, and the findings were inconclusive.^{20,26,28,41,56,61,62}

Discussion

This review provides evidence for factors to include in a social–ecological model of recess physical activity behavior but also highlights areas lacking evidence. Forty-four variables were identified across the four levels of the framework, although only 36% had been investigated four or more times. Only three studies^{38,46,48} examined correlates across all levels of the model simultaneously. The majority of variables identified were at the individual and physical environmental levels of the model, although only three consistent associations were found for gender, unfixed equipment, and facility provision. Few studies examined social influences, despite recess providing an

opportunity for youth to engage in social interactions relatively free of adult constraints.⁶⁸

Potentially important factors such as composition of social groups (e.g., single or mixed gender) have not been investigated; this may be somewhat explained by the many child studies^{25,38,40,46–48} reviewed with a focus on teacher supervision and management. In addition, further research should investigate correlates of children's and adolescents' recess physical activity for variables that were not investigated frequently but indicated positive associations with physical activity. Identifying recess physical activity correlates may help inform future intervention development and school programming, because although recess interventions have generated positive results, reported changes are small and often focus on the short term.^{30,39,64}

Some limitations of included studies warrant attention. The majority of the research in this area has examined small samples, utilizing cross-sectional study designs. A meta-analysis of the data may provide more information but is difficult given the limited number of studies that report effect sizes and the lack of consistency in correlates assessed. A range of physical activity measures also have been used, which may influence the associations identified. Although the majority of child studies have used objective measures of physical activity (particularly accelerometry and direct observation), different accelerometer cut-points and observation systems may have influenced the strength of associations observed. The majority of adolescent studies used self-report measures and although face and content validity of the measures were documented in some studies,^{15,53,54} further research using objective measures is needed to determine adolescents' recess physical activity levels.

Arguably the main weakness in the evidence is at the organizational/policy level, where only two variables have been investigated four or more times and the associations for recess duration and differences between recess periods and physical education were inconclusive. This weakness is concerning, especially given the increasing trend to reduce the frequency and/or duration of recess.⁶⁹ The way recess is defined may explain the lack of consistent findings, as several studies included time spent eating at lunchtime and thus a substantial component of compulsory sedentary behavior.^{37,40} Moreover, some studies reported total daily recess duration,^{35,60} whereas others examined the impact of individual recess periods,^{18,31} which may have influenced results.

Little research has documented the effect of written recess policies on recess provision and physical activity levels, particularly in countries where daily recess is not mandatory (e.g., the U.S.).⁷⁰ However, mandatory recess periods may be one approach to influencing children's

physical activity at the organizational/policy level.⁵¹ Schools are encouraged to have a written school physical activity programming policy that includes recess, which may contribute to stronger policies and programs at the school level.⁷⁰ This also may encourage schools that are contemplating the removal of recess to consider the benefits of retaining recess on not only social, emotional, physical, and cognitive development^{68,71} but also classroom behavior and academic performance.⁷¹ Further, adopting written school physical activity programming policies may help to not only promote more consistent scheduling of recess⁷⁰ but also contribute to children and adolescents' daily physical activity levels, as a substantial proportion of their recommended daily physical activity can be accumulated during this time.⁷²

Higher perceived encouragement from parents, peers, and the school as a whole was associated with higher self-reported physical activity levels during recess periods, particularly in adolescents.⁵³ It was surprising, however, that perceived parental encouragement influenced physical activity levels in younger adolescents, given the lack of involvement of parents have during this time. This result may be explained partly by the measures used, as the adolescents were not asked specifically about parental encouragement for physical activity and sport during school recess.⁵³ Interestingly, parental encouragement may benefit physical activity levels across a range of contexts, even in those where parental involvement is minimal.

Overall facility provision (i.e., sum of facilities available) was positively associated with physical activity.^{15,32,50,51} The provision of outdoor (e.g., sports fields, ball areas)¹⁵ and indoor spaces (e.g., gyms)⁵⁰ during recess periods was not associated with physical activity when they were investigated as individual facilities. These facilities may be associated with physical activity during other noncurricular periods, such as before or after school, although this was not examined in this review. Increasing access to different facilities during recess and/or lunchtime at school may benefit youth physical activity although single spaces may not be effective on their own. This supports research that has demonstrated that providing access to a range of spaces and facilities may stimulate physical activity by increasing a sense of choice and providing supportive environments that facilitate active behaviors.^{37,73,74} Future research should examine whether increasing access to school facilities during recess periods increases physical activity levels in children and adolescents.

A positive association was found between unfixed equipment and recess physical activity in children. No studies examined this association in adolescents. These results contrast with previous reviews^{10,11} of correlates

that reported no or inconclusive associations between the availability of toys/equipment and overall physical activity. It is possible that unfixed equipment may affect physical activity participation in specific contexts, rather than overall daily physical activity.

In the present study, there was no clear indication as to whether unfixed equipment increased MPA, VPA, or MVPA, which may be dependent on the type and amount of equipment provided. For example, Zask and colleagues⁴⁸ reported that the ratio of balls to children had an impact on VPA, whereas others used a dichotomous variable to identify the presence or absence of equipment and found positive associations with both MPA and VPA.^{25,38,40,47} Further research is needed to determine whether specific types of equipment, or the overall availability of unfixed equipment, are associated with higher levels of physical activity. In elementary school settings, providing games equipment has been found to increase children's physical activity levels.⁶⁴ Examining whether the provision of unfixed equipment is a suitable strategy for increasing adolescents' recess physical activity is warranted.

This review found that boys are more physically active than girls during recess, supporting previous reviews^{10–12} of preschool, childhood, and adolescence correlates. Boys view recess as an opportunity to play competitive games that often dominate the available space.^{75,76} Girls may view recess as an opportunity to socialize with friends.^{75,77} However, behaviors during recess have been studied primarily among children, and there are limited data concerning adolescents. There is a need to establish why adolescent boys are more active than girls during recess, as this will inform strategies to promote physical activity in this age group. Future research should examine the correlates of boys and girls physical activity separately. Identifying modifiable variables that differ by gender could be critical in the development of activity promotion strategies, and may help inform evidence-based practice and policies designed to increase physical activity in primary and secondary school children.

Conclusion

There is currently a dearth of literature concerning correlates of physical activity during recess periods, particularly in adolescents. Despite the paucity of associations identified, it is recommended that schools increase overall facility provision, provide unfixed equipment, and identify methods to increase social support, particularly by peers, to benefit children's and adolescents' physical activity during recess.

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Appendix

Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.amepre.2012.05.019>.